

1. a. No. $\mathbf{0} \notin \{x \in \mathbb{R}^2 : x_1 + x_2 = 1\}$ since $0 + 0 \neq 1$.

h. Yes. We can show this by proving that the set is $\text{Span}((2, 1, 1), (1, 2, -1))$. $s \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} + t \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$ gives a set

of vectors that is $\text{Span}((2, 1, 1), (1, 2, -1))$. So we need only show that $\begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix} \in \text{Span}((2, 1, 1), (1, 2, -1))$.

So let $s = 2$ and $t = -1$ which gives $(3, 0, 1) = 2(2, 1, 1) - 1(1, 2, -1)$. Thus the set is $\text{Span}((2, 1, 1), (1, 2, -1))$.

2. a. No. To show this we need to find the rank of the column matrix associated with the given span. Taking $\text{Span}((1, 1, 1), (1, 2, 2))$,

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 2 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 0 & 0 \end{bmatrix}$$

So $\text{rank}(A) = 2 < 3$ and $\text{Span}((1, 1, 1), (1, 2, 2)) \neq \mathbb{R}^3$.

- d. Yes. As above we find the rank of the associated column matrix.

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ -1 & 1 & 5 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

So $\text{rank}(A) = 3$ and $\text{Span}((1, 0, -1), (2, 1, 1), (0, 1, 5)) = \mathbb{R}^3$.

5. $V = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = B\mathbf{x}\}$

Proof. i. (Need to show that $\mathbf{0} \in V$.) $\mathbf{0} \in \mathbb{R}^3$ and $A\mathbf{0} = 0 = B\mathbf{0}$ so then $\mathbf{0} \in V$.

ii. (Need to show that if $\mathbf{v} \in V$ and $c \in \mathbb{R}$ then $c\mathbf{v} \in V$.) Let $c \in \mathbb{R}$ and $\mathbf{v} \in V$. $A(c\mathbf{v}) = cA\mathbf{v} = cB\mathbf{v} = B(c\mathbf{v})$ so $c\mathbf{v} \in V$.

iii. (Need to show that if $\mathbf{v}, \mathbf{w} \in V$ then $\mathbf{v} + \mathbf{w} \in V$.) Let $\mathbf{v}, \mathbf{w} \in V$. $A(\mathbf{v} + \mathbf{w}) = A\mathbf{v} + A\mathbf{w} = B\mathbf{v} + B\mathbf{w} = B(\mathbf{v} + \mathbf{w})$ so $\mathbf{v} + \mathbf{w} \in V$.

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